

Sec. 3.1: Sets and subsets

Definition 1.

- A **set** is a collection of objects.
- The objects in a set are called **elements** or **members** of the set.

Notation:

- Roster notation: $A = \{3, 1, 2\}$, $B = \{\text{ice cream, coffee, donut}\}$, $C = \{1, 2, 3, 3\}$, $D = \{x, y, \{1\}\}$.
- Set builder notation: $\{x : P(x)\}$, $\{(\text{expression}) : (\text{expression}) \text{ has some properties}\}$.
- “ $\in$ ”:
- “ $\notin$ ”:

Common mathematical sets:

- The set of **natural numbers**:
- The set of **integers**:
- The set of **rational numbers**:
- The set of **real numbers**:
- The set of **complex numbers**:
- The set of **even integers**:
- The set of **odd integers**:
- The set of **irrational numbers**:

Relationship of $\mathbb{C}, \mathbb{N}, \mathbb{Q}, \mathbb{R}, \mathbb{Z}$:

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. We will define “ $\subset$ ” symbol (proper subset) in the next page.
(Exercise: Show that each “ $\neq$ ” is true (find examples!))

Definition 2.

- The set with no elements is called the **empty set** (or **null set**), denoted by or .
- The **universal set**, denoted by , is a set that contains all elements mentioned in a particular context.

Definition 3. Let A be a set.

- A is **finite** if either
 - (a) $A = \emptyset$.
 - (b) Elements in A can be numbered 1 through n for some positive integer n .
- The **cardinality** of finite set A , written , is
 - (a) $|A| = \text{}$, if $A = \emptyset$.
 - (b) $|A| = \text{}$, if elements in A can be numbered 1 through n for some positive integer n .
- A is **infinite** if it is not finite.

Example 1. Roster notation, set builder notation, and cardinality.

- $A = \{x \in \mathbb{R} : x^2 - 3x = 0\} =$
- $B = \{r \in \mathbb{Q} : r^2 = 2\} =$
- $C = \{4, 8, 12, 16, 20, \dots\} =$

Definition 4. Two sets A and B are **equal**, written as , if and only if they have exactly the same elements.

Note: This includes the case that neither set contains any element.

Definition 5. Let A and B be two sets.

- A is a **subset** of B , written , if and only if every element of A is also an element of B . More precisely (useful in proofs!):
- If $A \subseteq B$ but $A \neq B$, then A is a **proper subset** of B , written .
- Alternative definition of $A = B$ (useful in proofs!):

- Does the order of the list, or the number of times an element is listed affect the set?
- Does adding or removing any brackets affect the set?

Example 2. The following statements are true.

1. $a \in \{a, b, \{c\}\}$
2. $b \in \{a, b, \{c\}\}$
3. $c \notin \{a, b, \{c\}\}$
4. $\{c\} \in \{a, b, \{c\}\}$
5. $\{a, \{c\}\} \subseteq \{a, b, \{c\}\}$
6. $\{a, \{c\}\} \subset \{a, b, \{c\}\}$
7. $\{a, b, c\} \not\subseteq \{a, b, \{c\}\}$

Example 3. For each of the following blanks, which symbol(s) can be used?

	$\subseteq$ or $\not\subseteq$?	$\subset$?	$\in$ or $\notin$?
$\{1, 2\}$ $\{1, 2, 3, \{1, 2, 3\}\}$			
$\{\{1, 2\}\}$ $\{1, 2, 3, \{1, 2, 3\}\}$			
$\{1, 2\}$ $\{1, 2, \{1, 2\}\}$			
$\{\{1, 2\}\}$ $\{1, 2, \{1, 2\}\}$			

Theorem. For any set A , $\subseteq A$ and $\subseteq A$.

Summary - fill in the blanks. Answers will be posted on Canvas.

- A **set** is a collection of objects. The objects are called or of the set.
- If $A = \{1, 2, 3\}$, then 1 A , and 4 A .
- The symbol $\emptyset$ denotes .
- The **cardinality** of a finite set A , written , is .
- A is a **subset** of B , written , if and only if .
- A is a **proper subset** of B , written , if and only if .
- Sets A and B are equal if and only if A B and B A .
- For any set A , $\subseteq A$ and $\subseteq A$.
- Fill in the blanks with $\mathbb{C}, \mathbb{N}, \mathbb{Q}, \mathbb{R}, \mathbb{Z}$: $\subset$ $\subset$ $\subset$ $\subset$.